

WISMITHA PEIRIS

Electronic & Embedded Engineer | Real Time AI Pipelines (STT, LLM, TTS) · Robotics · Embedded ML · PCB Design

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Portfolio: wismithapeiris.github.io | Live product: eventplanners.app | Open to relocation

PROFESSIONAL SUMMARY

Electronic and Embedded Engineer who builds real time AI systems that ship. At Figmentum Robotics I completely built the conversational AI of a children’s educational robot: speech to text, language models and text to speech engineered end to end for ultra low latency on H100 and H200 GPU servers rented from JarvisLabs.ai, together with the test bench platforms that prove every model choice against ground truth. Earlier I built an industrial predictive maintenance platform end to end, from custom PCB to cloud ML, delivered a PLC production machine running since 2023, and shipped a full Android SaaS live on Google Play. BSc Hons in Electrical and Electronic Engineering (University of Peradeniya); MSc in progress at Anglia Ruskin University, UK. I learn whatever the problem demands, measure instead of guessing, and deliver working systems from hardware bring up to production.

TECHNICAL SKILLS

Programming	Python (primary), C, C++, Kotlin, TypeScript, Bash
Real Time AI Pipelines	STT, LLM and TTS integration; wake word and VAD; push to talk system design; ultra low latency streaming; model serving with vLLM and NVIDIA NeMo; prompt engineering and child safety design
Model Evaluation	ground truth benchmarking, WER and CER scoring, OCR accuracy analysis, latency and cost measurement, automated PDF and HTML reporting
Computer Vision	OpenCV, PaddleOCR, vision language models, handwriting OCR, OAK D, Raspberry Pi Camera Module 3, IMX500
GPU, Cloud & Backend	rented H100 and H200 GPUs (JarvisLabs.ai), Docker, nginx, FastAPI, WebSockets, AWS (S3, Parquet), Firebase (Firestore, Cloud Functions, FCM), Cloudflare CDN, Streamlit
Hardware & PCB	Altium Designer, KiCad, EasyEDA; schematic to layout, hardware bring up, failure analysis, JTAG, oscilloscope and DMM, soldering, electrical panel design
Embedded Systems	Raspberry Pi 4 and 5, ESP32 (ESP IDF), Arduino Mega 2560, Jetson AGX; ADC, I2C, SPI, UART, Modbus; real time multi threaded acquisition
Robotics & Controls	Model Predictive Control, LiDAR mapping, path planning, PID control, sensor fusion, autonomous navigation
Mobile (Android)	Kotlin, Jetpack Compose, MVVM, Google Play Billing, WhatsApp deep linking, Chromecast integration
Industrial Automation	PLC programming (Siemens, Delta), HMI design, pneumatic integration, safety interlocks, transformer fundamentals (amorphous core), battery backup power switching
Design & Simulation	SolidWorks, SolidWorks Electrical, AutoCAD, Proteus

PROFESSIONAL EXPERIENCE

Prototype Engineer · Figmentum Robotics LLC (USA, Remote) Sep 2025 to Present

Built the complete conversational AI, vision and model validation stack of a children’s educational robot.

- Built the robot’s first conversational pipeline in Python: wake word, VAD, speech to text, an LLM with emotion tags and emotion aware text to speech, plus highly attractive projected math and spelling games with camera based handwriting OCR and step by step tutoring.
- Measured through real child testing that voice activity detection sometimes added 2 to 5 seconds per answer, then designed and built a push to talk platform: a Raspberry Pi 5 streaming voice over one warm secure WebSocket to H100 and H200 GPU servers rented from JarvisLabs.ai, tuned around one number, button release to first sound.
- Researched and served child appropriate open models for every stage (Parakeet TDT, Canary, Kid Whisper, Kyutai; Qwen3 VL, Llama 3.1, GPT OSS 20B, Phi 4 Mini, Mistral NeMo; Chatterbox Turbo, Kokoro, Maya1, Dia, Orpheus, Piper), swappable per turn with fallbacks to find the best combination.
- Built the company’s test bench validation platform (FastAPI backend, React frontend): an STT tab that validates models against ground truth across seven strict sections with professional PDF reports, and a vision tab that benchmarks more than 25 handwriting OCR models with honest per stage latency.
- Engineered a Raspberry Pi capture station (Camera Module 3, fixed rig) with full sensor preview, autofocus confirmation and stability and sharpness gates for perfectly repeatable OCR dataset photos.
- Privacy and safety by design: child audio never leaves the rented servers, an always on child system prompt shapes every answer, and an optional guard model checks each clause before it is spoken.
- Evaluated edge controllers and accelerators for latency, thermals and power to select on device deployment hardware.

Electronic Project Engineer · Hyphen Holdings (Pvt) Ltd, Horana Nov 2024 to Aug 2025

Built the full predictive maintenance hardware platform end to end: sensors to edge to ML to cloud.

- Programmed industrial vibration, current and temperature sensors for high speed real time acquisition on Raspberry Pi 4 (Python), with sampling and synchronization designed for harsh factory conditions.
- Designed the complete system architecture spanning sensor interfacing, edge processing, ML inference and cloud storage.
- Designed a custom PCB from schematic to layout for a battery backup module with high speed power switching that guarantees uninterrupted operation through line power loss.
- Trained and optimized ML anomaly and fault detection models on time series data, including FFT spectrum analysis, bearing fault frequencies and ISO 10816 severity, with adaptive per machine baselines.

- Implemented AWS storage pipelines using Parquet and columnar compression to cut storage cost and speed retrieval, feeding a React dashboard deployed with Docker and Nginx.

Trainee Robotics & Automation Engineer · Midas Safety (Pvt) Ltd Jul 2023 to Oct 2023

Biyagama and Awissawella. Industrial automation and PLC programming on glove production lines.

- Designed and built a production heat transfer machine for glove printing: Delta PLC programming, HMI design, electrical panel design, wiring and pneumatic integration; running on the Awissawella floor since 2023.
- Designed a secondary electrical control panel synchronized with the main panel that measurably reduced the glove defect rate by improving process control.
- Conducted a plant wide electrical panel survey identifying unsafe and non compliant conditions across the facility.

Trainee Electrical Engineer · LTL Transformers (Pvt) Ltd, Moratuwa May 2022 to Aug 2022

- Conducted amorphous core transformer R&D, evaluating core materials to reduce energy losses in distribution transformers.

PROJECTS

Children Interaction Robot: API Voice Pipelines to a Push to Talk Platform · Fimentum Robotics LLC · 2025 to 2026

The complete conversational brain of an educational robot. Built the API era first (wake word, STT, LLM, emotion aware TTS, projected games with camera OCR), then, after measuring 2 to 5 second VAD delays, a push to talk platform: the Pi 5 as a thin device streaming to H100 and H200 GPU servers rented from JarvisLabs.ai over one warm WebSocket, running researched open STT, LLM and TTS models tuned for button release to first sound, with per turn barge in handling and full child privacy.

STT Accuracy Test Bench & Validation Platform · Fimentum Robotics LLC · 2026

Single port web app (FastAPI, React) validating speech to text models against ground truth across seven sections: WER accuracy, pauses and disfluency, noise robustness, end of turn, STT latency, spelling CER and projected cost, with ship blocker gating and professional PDF reports. Runs GPT 4o Transcribe beside Parakeet TDT, Canary, Kid Whisper, Kyutai and Whisper large v3 turbo served from a rented H100 GPU through an SSH tunnel.

OCR Simulation & Benchmarking Platform with Pi Capture Station · Fimentum Robotics LLC · 2026

Vision test bench for more than 25 handwriting OCR models (Nemotron, Ministral, Molmo2, TrOCR, Llama 3.2 Vision, PaddleOCR VL, Baidu Unlimited OCR, GPT and Gemini families, Google Cloud Vision) on a rented H100 class GPU: three stage pipeline of page geometry, PaddleOCR word detection with orientation, and per crop reading; verbatim outputs, per stage latency and around four hundred automated tests. Paired with a Raspberry Pi capture station for perfectly repeatable dataset photos, planned with a camera position simulator I built that models every Pi camera module at any height, tilt and pan and computes the resolution landing on the page before the rig was physically built.

EventPlanners Digital Invitation SaaS · WellSmart Automations · live at eventplanners.app · 2024 to 2026

Complete event SaaS built alone: Kotlin and Jetpack Compose app with WhatsApp first delivery (per guest personalization, bulk send, broadcast mode), a dynamic single link that changes across the event lifecycle, AI writer via Cloud Functions, QR check in, seating, budget PDF reports, post event thank you cards with gallery photo attachments, a Cloudflare CDN with HMAC signed URLs cutting Firebase egress by over 95%, and Google Play Billing tiers.

Autonomous Vehicle Platform: Vision, LiDAR & Real Time Web Dashboard · Independent Engineering Project · 2026

Raspberry Pi 5 platform with dual IMX500 AI cameras, TF Luna LiDAR braking, servo steering and motor control; an eight state finite state machine, computer vision lane detection, PID steering with anti windup, sensor fusion, and a Flask and Socket.IO dashboard with dual live feeds, telemetry, live tuning and emergency stop, plus a hardware free simulator.

Predictive Maintenance Hardware Platform · Hyphen Holdings (Pvt) Ltd · 2024 to 2026

Industrial Raspberry Pi platform with vibration, temperature and current sensors, a custom battery backup PCB, ML fault detection on time series data, an AWS Parquet pipeline and a React dashboard behind Docker and Nginx.

Object Avoidance Robot using Model Predictive Control (MPC) · University of Peradeniya · 2024

CAD designed structure, LiDAR mapping and obstacle detection, and MPC path planning fully onboard a Jetson AGX for reliable autonomous navigation.

Heat Transfer Machine: Glove Printing Automation · Midas Safety, Awissawella · 2023

Delta PLC programming, HMI design and pneumatic integration for a production deployed glove printing machine, on the factory floor since 2023.

Smart Home Automation System (ESP32) · University of Peradeniya · 2023

Custom KiCad PCB and ESP IDF firmware in C: scene and schedule engines, OTA updates, Google Home and Alexa over MQTT, touchscreen control, and a local first design that works through internet outages.

Double Drive Design for Optimizing Glove Manufacturing · Midas Safety, Awissawella · 2023

A new electrical control panel synchronized with the main panel, reducing the glove defect rate by enhancing process control and production efficiency.

Vending Machine Prototype · University of Peradeniya · 2022

Working vending machine prototype with CAD and an Arduino Mega 2560 control system: coin acceptance, item selection, dispensing and status display.

EDUCATION, MEMBERSHIP & ACTIVITIES

MSc, Electronic and Electrical Engineering · Anglia Ruskin University, UK 2025 to 2026

BSc Eng. (Hons), Electrical and Electronic Engineering · University of Peradeniya, Sri Lanka 2020 to 2024

Programming for Everybody (Python) · University of Michigan (online) 2020

C for Everyone: Programming Fundamentals · University of California (online) 2020

Diploma in Western Music · Trinity College London Examinations Completed 2018

Associate Member, Institute of Engineers Sri Lanka (IESL) | Member, University of Peradeniya Athletics Team (2020 to 2024)